Online Resource 4

Audit reliability

Supplementary material to: *How the Classical Baseline Shapes Machine-Learning Speed Claims in Topology Optimization.* Shaoliang Yang and Jun Wang, Santa Clara University. Corresponding author: Jun Wang, jwang22@scu.edu. Computers & Structures.

Reading guide. R1–R4 assess quality matching, classical efficiency, offline cost and verification of returned designs. A speed-stratum article has a primary speed or cost comparison; a quality-stratum article compares design quality only. CI, confidence interval; ML/AI, machine learning/artificial intelligence; TO, topology optimization; FE/FEM/FEA, finite-element method/analysis; CNN/DNN/ANN/NN, convolutional/deep/artificial/neural network; CPU/GPU, central/graphics processing unit; DOF, degree of freedom; SIMP, solid isotropic material with penalization; OC, optimality criteria; MMA, method of moving asymptotes; MSE/RMS/MAE/MAPE, mean squared/root mean square/mean absolute/mean absolute percentage error; VAE, variational autoencoder; CMAME, Computer Methods in Applied Mechanics and Engineering. Counts use the applicable denominator printed in each row. Binomial intervals use the exact method of Clopper and Pearson (1934) and describe sampling of coded judgments. Screening reason codes (E- and I-prefixed) and rule markers are defined in Online Resource 1; API, application programming interface. Article-specific labels in quoted coder notes refer to the cited article.

Reader's note. Quoted evidence and marker texts come from the coders' working records, reproduced as written; in the access notes of section 7, 'this agent' is the model coder that wrote the note, and [recalled] marks a statement made from memory. Evaluative terms describe how reported comparisons score against the audit rules of Online Resource 1, not the merit of the articles or the intent of their authors.

1. Verdicts by publication year

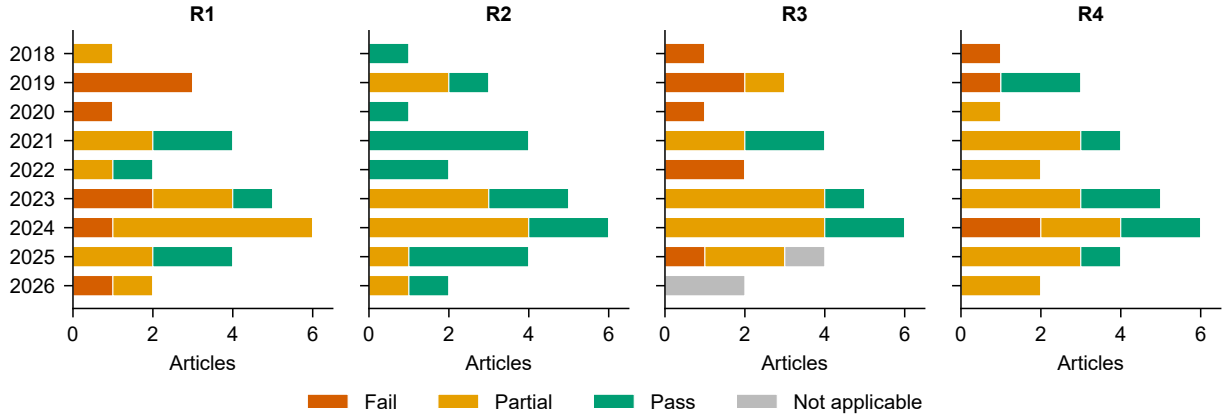


Fig. OR4.1 Rule verdicts of the 28 speed-stratum articles by publication year. R1, quality matching; R2, classical efficiency; R3, offline-cost accounting; R4, verification of returned designs. Source: `audit/coding/records/*.json`; plotted values in `or4_fig1_verdicts_by_year.csv`.

2. Scale sensitivity

Reported rates count only *fail*. The sensitivity analysis counts *partial* as a failure too (`c1_rules.json`, `c1_rules_sensitivity.json`).

Stratum	Rule	Applicable	Fail only (95 % CI)	Fail or partial (95 % CI)
speed	R1	28	8 (29 %; 13–49 %)	22 (79 %; 59–92 %)
speed	R2	28	0 (0 %; 0–12 %)	11 (39 %; 22–59 %)
speed	R3	25	7 (28 %; 12–49 %)	20 (80 %; 59–93 %)
speed	R4	28	4 (14 %; 4–33 %)	20 (71 %; 51–87 %)
speed	weak baseline (R1 or R2)	28	8 (29 %; 13–49 %)	24 (86 %; 67–96 %)
speed	any rule	28	13 (46 %; 28–66 %)	27 (96 %; 82–100 %)

Stratum	Rule	Applicable	Fail only (95 % CI)	Fail or partial (95 % CI)
quality	R1	29	1 (3 %; 0–18 %)	27 (93 %; 77–99 %)
quality	R2	3	2 (67 %; 9–99 %)	2 (67 %; 9–99 %)
quality	R3	0	not applicable	not applicable
quality	R4	29	5 (17 %; 6–36 %)	26 (90 %; 73–98 %)
quality	weak baseline (R1 or R2)	29	3 (10 %; 2–27 %)	27 (93 %; 77–99 %)
quality	any rule	29	8 (28 %; 13–47 %)	29 (100 %; 88–100 %)

3. Agreement between the two blind codings

Cohen's κ, a chance-adjusted agreement score (Cohen, 1960), with bootstrap 95 % intervals over the paired records (110 for inclusion and stratum, 82 for R1–R4; `agreement.json`, pre-registered). The post hoc rows restrict to the records that both codings placed in the speed stratum (21 pairs; `agreement_posthoc.json`; descriptive only).

Subset	Item	Pairs	Raw agreement	κ	95 % CI
all records	included	110	88 %	0.62	0.42 to 0.80
all records	stratum	110	70 %	0.53	0.40 to 0.65
all records	R1	82	55 %	0.35	0.21 to 0.49
all records	R2	82	56 %	0.28	0.12 to 0.43
all records	R3	82	83 %	0.66	0.49 to 0.80
all records	R4	82	70 %	0.56	0.42 to 0.70
speed stratum (post hoc)	R1	21	57 %	0.33	0.00 to 0.64
speed stratum (post hoc)	R2	21	52 %	0.14	-0.19 to 0.49
speed stratum (post hoc)	R3	21	90 %	0.86	0.63 to 1.00
speed stratum (post hoc)	R4	21	81 %	0.70	0.38 to 0.93

4. Results per coding

The per-coding rates are post hoc; the 'final' row is the adjudicated coding that the paper reports. The G1 band is the framing that the protocol assigns to the G1 value (Online Resource 1). G1 is the pre-registered decision value (first 20 speed-stratum records); the sample columns use every speed-stratum record of that coding (`coder_sensitivity.json`).

Coding	G1: weak baseline	G1 band	Sample: weak baseline
first	6 of 20 (30 %; 12–54 %)	specialist-journal framing	12 of 26 (46 %; 27–67 %)
second	5 of 20 (25 %; 9–49 %)	specialist-journal framing	10 of 38 (26 %; 13–43 %)
final	5 of 20 (25 %; 9–49 %)	specialist-journal framing	8 of 28 (29 %; 13–49 %)

5. Blind re-screen of excluded records

A second model re-screened a seeded random sample of 188 title-and-abstract excludes blind to the first decision (`rescreen.json`; not adjudicated). It included 17 (9 %; 5–14 %), all as unclear from the abstract; 15 of them were eligible at the draw (8 %; 5–13 %). Scaled to all excludes, that is about 170 records (101–265). Both passes excluded 171 records, 150 with the same code.

Record	Title	First code	Re-screen	Re-screen rationale
P00005	Towards end-to-end optimization in multimaterial 3D printing	E-FORWARD	I-UNCLEAR	NN surrogate constitutive model embedded in FE-based topology optimization for multimaterial printing; no classical-method comparison stated.
P00069	Multiscale topology optimization of functionally graded lattice structures based on physics-augmented neural network material models	E-NOCLAIM	I-UNCLEAR	Physics-augmented neural-network material model embedded in multiscale lattice topology optimization; abstract states no explicit comparison to a classical baseline.

Record	Title	First code	Re-screen	Re-screen rationale
P00110	Data-driven Topology Optimization (DDTO) for Three-dimensional Continuum Structures	E-FORWARD	I-UNCLEAR	Neural-network constitutive surrogate embedded in explicit topology optimization for materials without known constitutive laws; no classical comparison stated.
P00390	Quantum topology optimization of ground structures using noisy intermediate-scale quantum devices	E-NOML	I-UNCLEAR	Variational quantum algorithm optimizes truss ground-structure topology; unclear if it counts as a learned model, no stated classical comparison.
P00586	Topology optimisation for design and additive manufacturing of functionally graded lattice structures using derivative-aware machine learning algorithms	E-NOCLAIM	I-UNCLEAR	Derivative-aware ML algorithm for multiscale lattice topology optimization in additive manufacturing; no explicit classical-method comparison stated in abstract.
P00869	Structural optimization of lattice structures using deep neural networks as geometry representation	E-NOCLAIM	I-UNCLEAR	DeepSDF neural-network geometry representation for lattice structural optimization, contrasted with classical TO methods; no explicit performance comparison stated.
P00972	A latent space approach to multi-material topology optimization	E-NOCLAIM	I-UNCLEAR	VAE-built material latent space coupled with gradient-based multi-material topology optimization; no explicit classical-method comparison stated in abstract.
P01017	Data-driven multiscale nonlinear topology optimization of functionally graded soft composites under large deformations	E-NOCLAIM	I-UNCLEAR	Neural-network material surrogate embedded in multiscale nonlinear topology optimization of soft composites; no explicit classical-method comparison stated.
P01358	Topology optimization of open radiation shielding structures	E-NOML	I-UNCLEAR	Empty abstract; title confirms topology optimization of shielding structures but not whether a learned model or comparison is used.
P01375	Functionally graded structure design for magnetic field applications	E-DOMAIN	I-UNCLEAR	Abstract empty; title/venue (CMAME) are compatible with a structural design for a magnetic-field environment, but domain and ML use are genuinely unclear.
P01432	A Human-Aided Topology Optimization Method Based on Moving Morphable Spline Components	E-NOML	I-UNCLEAR	Empty abstract; title confirms moving-morphable-component topology optimization in solid mechanics, but learned-model use and comparison are unstated.
P01585	An Efficient Lightweight Design Framework for Movable Platen in Two-Platen Injection Molding Machine Integrating Machine Learning and Multi-objective Optimization Algorithm	E-SCOPE	I-UNCLEAR	Empty abstract; title confirms ML-integrated multi-objective lightweight structural design of a machine part, plausibly topology optimization.
P02003	Physics-Informed Generative Design for Nonlinear Topology Optimization of Aerospace Bracket Assemblies Compliant with ASTM Standards	E-MLONLY	I-UNCLEAR	Implicit-neural-field nonlinear topology optimization of aerospace brackets; compares to a "generative baseline" of unclear classical vs. ML status.
P02211	A machine learning-based surrogate model for optimization of truss structures with geometrically nonlinear behavior	E-SCOPE	I-UNCLEAR	Title indicates an ML surrogate model accelerating optimization of geometrically nonlinear truss structures; abstract empty, explicit classical-method comparison not confirmed.
P02212	A novel deep unsupervised learning-based framework for optimization of truss structures	E-SCOPE	I-UNCLEAR	Empty abstract; title confirms a deep unsupervised-learning framework for truss-structure optimization, in-scope method and physics, comparison unstated.
P02282	Neural Optimization Machine for Design With Neural Network Based Objectives and Constraints	E-SCOPE	I-UNCLEAR	Physics-guided neural-network-based optimizer applied to AM fatigue-property design; whether it is topology optimization and any classical comparison are unstated.
P02535	Toplogical optimization of structures using Fourier representations	E-NOML	I-UNCLEAR	Title/venue compatible with either classical Fourier-series topology-optimization parameterization or a neural Fourier-feature method; abstract empty, ML use unclear.

6. Adjudication decisions

66 of 110 records had at least one disagreement between the blind codings; the adjudicator settled 247 items from the article text (R1: 46, R2: 48, R3: 28, R4: 34, access: 14, claimed speed-up: 22, decision: 13, full-text exclusion code: 11, stratum: 31). All 80 adjudicated items of the 28 speed-stratum records are listed below with the adjudicator's recorded reason; the others are in the spreadsheet of Online Resource 3 (column adjudicated items).

Record	Item	First	Second	Decision	Reason
P00080	R1	fail	partial	partial	The timed classical scenarios are projections with no design (Sec. 3.3, pp. 10-11), but the same mid-cantilever design is compared on quality with the classical 88-line and infill designs (61.05 and 76.86 vs 68.3 predicted, 74.6 on FE re-analysis at volume 0.65; Sec. 3.2.2, pp. 8-9). So the 'no quality comparison' fail marker is not met (as in P00140), and the unmatched volume and assumed 300 iterations are the §5.3 R1 partial case.
P00080	R2	partial	pass	partial	The brute-force projection rests on one iteration of 16.62 min at 1600 x 800 elements (about 2.56 million DOFs [derived]) on a laptop, with code and solver unstated (Sec. 3.3, p. 10). That time is far above what the authors' own homogenization cost (0.132 s at 120 x 120) implies even with superlinear scaling [inferred], so the implementation looks inefficient, but only an R2(e) re-test can compare it. The 88-line code that the second coder cites is named only for the Sec. 3.2.2 quality design; the code of the timed iteration is not stated.
P00080	R4	partial	pass	partial	The mid-cantilever and heel-bone designs are FE re-analysed (Sec. 3.2.2, p. 9; Sec. 3.2.4, p. 10), but the 0.65 volume against the 0.6 bound is presented as a 6.1% error rather than as a violation, the heel-bone volumes appear only as errors below 10%, the tensile-bar and parameter-study compliances are not stated as FE-verified (Figs. 14, 17-19), and connectivity failures are not counted: the §5.3 R4 partial case.
P00080	claimed speed-up	2200	174.1	2200	No factor is stated. The per-run ratio against the concurrent-homogenization projection, 528 / 0.24 = 2,200 (Sec. 3.3, pp. 10-11) [derived], uses the comparator the conclusion names (Sec. 4, p. 11) and is the lower of the two per-run ratios; the project keeps data and training costs out of the factor (as in P00176 and P00183) and handles them under R3. The second coder's 174.1 divides the brute-force projection by the one-time plus per-run time, which stays in the basis as an alternative.
P00140	R1	fail	partial	partial	Quality is reported, but only against the homogenization reference (Tables 2-5; abstract 7-25%). The timed comparator [16] is a cited time on other hardware whose design quality is never given (Sec. 4.2.1, p. 18). The text establishes neither fail condition: the CNN output is not shown worse than [16]'s output, and a quality comparison does exist. What remains is the §5.3 R1 partial case, quality reported with loose matching.
P00140	R2	pass	partial	pass	For the primary comparator [16], the text establishes none of (a)-(e): the abstract gives no baseline figure, both timings come from laptops (this work: a single Intel Core i7 CPU, Sec. 4.1), and [16]'s iterations and implementation are not given (Sec. 4.2.1, p. 18). The second coder's partial concerns top88, which is the alternative comparator, and unsure means pass.
P00140	claimed speed-up	5	7.5	5	The article states only a range, a factor of 5 to 10 (Sec. 4.2.1, p. 18; Conclusions, p. 22), so the conservative lower bound is entered. The derived per-size ratios, 10/1.3 = 7.7 and 10/2.2 = 4.5, span the same range, and the second coder's 7.5 is the midpoint of the range, a coder construct.
P00149	R1	partial	pass	pass	The primary outcome is per stiffness evaluation, and the DNN is validated against the same FE homogenization it replaces ($R^2 \geq 0.996$ per component on 2,130 held-out samples, Sec. 5.1, p. 12), so the comparison sits within the ML method's reported error of the classical result (§5.3 R1 pass). The first coder's reasons (no FE^2 run, unstated RVE mesh, runtimes called qualitative, App. B, p. 23) concern the claim's scope, the baseline's cost and the timing protocol rather than quality matching; unsure means pass.
P00149	R3	partial	pass	partial	Training cost is reported (10 min, Table B.1, p. 23), but the cost of generating the 21,300 FEM-labelled samples (Sec. 5.1, p. 12) is never stated; it can only be inferred by multiplying the counts by the 5-min per-evaluation time, and the data were adapted from Kumar et al. 2020a (Table B.1 note). That is the §5.3 R3 partial case (data-generation cost missing), whatever the reading of the per-prediction footnote, and no break-even is given.

Record	Item	First	Second	Decision	Reason
P00150	R1	partial	pass	partial	PEN inference is timed against complete top88 runs whose stopping rule is not stated (Table 2, p. 15; Sec. 3.2, p. 16), which is the §5.3 R1 partial example; PEN's volume is only penalized, with deviations up to 0.010 in the six printed cases (Eq. 21, p. 9; Table 3, p. 17). Fail is not established, because no aggregate compliance gap is stated and the printed pairs are mixed (Fig. 15, p. 18).
P00150	R2	partial	pass	pass	The text establishes none of (a)-(e): no baseline figure in the abstract, 64×64 is far below 10^5 DOFs, and no iteration count is given (Sec. 3.2, p. 16; Table 5, p. 19). The timing hardware of both methods is unstated (Table 1, p. 12, is the training set-up), so a GPU-vs-CPU mismatch is only inferred, and the first coder itself was unsure; unsure means pass.
P00150	claimed speed-up	259	225	259	The version of record is open access and was read (mdpi-res.com PDF), so it is the coded version (PROTOCOL §4): it states 7.3 ms vs 1.9 s, 'roughly 259 times' (Sec. 3.2, p. 16). The second coder's 225 comes from arXiv v2 (8.4 ms, p. 12), an earlier version, because the mdpi.com page blocked it.
P00161	R1	fail	partial	fail	CNN inference is timed against complete SIMP runs (Table 1, p. 8), while the worst fifth of the test outputs, which include disconnected designs, has a mean compliance error of 103,872% and the volume misses its target by 9.70% on average (Table 2, p. 9); the conclusion's near-identity claim covers only the best 80% (Sec. 5, p. 11). A measurably worse output with no matching attempted is the §5.3 fail marker (as in P01217), not loose matching.
P00161	stratum	speed	quality	speed	The abstract has no numbers, and the conclusion holds two quantitative SIMP comparisons, the 2.75% compliance error and the 83% time cut (Sec. 5, p. 11). §5.1 rule 4 takes the time cut: the abstract's only explicit comparison with the classical code is speed (p. 1), the main text reports the time comparison first (Sec. 4.3, Table 1, p. 8), and on quality the article claims only near-parity with SIMP. This is the reading of adjudicated records P00183, P00890 and P02187.
P00176	R2	pass	partial	pass	Both methods ran on the same stated PC, i7-8700K plus GTX 1080 Ti (Sec. 5, p. 7), and the timed SIMP is the 88-line code (vectorized [recalled]) at 40×80, about 6.6×10^3 DOFs (Sec. 4.1, p. 4), so (c) cannot apply and no inefficiency is evident. The abstract gives no baseline figure (b), and no iteration count exists for (d) or (e). The unstated device per method is a reporting gap (hardware_per_method_stated = partial), not an established R2 case; unsure means pass.
P00181	R1	partial	fail	partial	Quality is reported, but only as voxel accuracy against the converged TopOpt design (96.2% binary, 79.7% RMS; Sec. 5.1, p. 12), with no compliance or volume, no equal-accuracy SIMP stop (Fig. 6, p. 9) and an unstated stopping rule (Sec. 4.1, p. 6): loose matching, the §5.3 R1 partial case. Fail needs a measurably worse output or no quality comparison; the voxel metrics are a quality comparison, and no design-level gap is measured, though the authors call the output sub-optimal (Sec. 6, p. 14). The image-only quality is penalized under R4 (fail).
P00183	R2	partial	pass	partial	The article's own text shows a custom gradient-descent augmented-Lagrangian solver that needs about 5000 FE analyses for a 4800-element 2D problem (Sec. 3.2, p. 6; Alg. 2, p. 17) and 10^2–10^3 s per topology (Sec. 5.4, p. 13), while the authors note MMA is also applicable (Appendix, p. 16). That is the §5.3 R2 partial case of an implementation that looks inefficient where no direct comparison is possible; (a) and (e) wait for the re-test, and (b)-(d) are not established.
P00183	claimed speed-up	10000	31623	10000	The article gives only orders of magnitude, 10^{-2} s vs 10^2 to 10^3 s per topology (Sec. 5.4, p. 13), so the derivable factor is a range of 10^4 to 10^5. The conservative lower bound is entered, as for other ranges in this audit (e.g. P00140); the second coder's 31,623 is the range's geometric mean, a coder construct.
P00549	claimed speed-up	18000	28200	18000	The primary outcome is the Conclusion's comparison (Sec. 4, p. 13): under 100 ms against approximately 0.5-1 h per FEM calculation, i.e. 18,000-36,000 [derived], and the project enters the lower bound of a stated range (as in adjudicated record P00140). The second coder's 28,200 comes from the 47 min worked example in Sec. 2.1 (p. 4), a main-text alternative that §5.1 ranks after the Conclusion.
P00550	access	no full text	full text	full text	The second coder's route worked for the adjudicator: Internet Archive snapshot 20250319061732 of the PDF on the corresponding author's former lab page, whose live URL now returns 404 while the author's new site shows a Cloudflare check (not bypassed). The file is the typeset version of record (CMAME 398, 115116; 24 pp.), read in full.

Record	Item	First	Second	Decision	Reason
P00550	stratum	None	speed	speed	The abstract (p. 1) claims up to 30 times faster than traditional topology optimization, a quantitative speed comparison with a classical method, so it is the primary outcome (§5.1 rule 1) and the stratum is speed.
P00550	R1	None	partial	partial	Both methods run the same 200-iteration schedule (Sec. 6, p. 14), and at 1.3 million elements Fig. 13 (p. 18) gives reference 15.4 against ML 15.6 and 15.7 (Fig. 12, p. 17). The headline 26.5x run (10.6 million elements, NB = 3; Fig. 16, p. 20) has no reference compliance, so quality is reported but matching is loose (§5.3 partial).
P00550	R2	None	pass	pass	Traditional TO and the ML framework share the machine, the in-house C++/CUDA FEA on a Titan Xp GPU and the iteration schedule (Sec. 6, p. 15; Sec. 6.4, p. 18), so (c) and (d) are not established. The solver type is unstated and (a)/(e) need a re-test, so unsure means pass.
P00550	R3	None	fail	fail	No training time and no data-generation cost appear anywhere: Sec. 6.1 (p. 15) calls the four training runs relatively cheap without figures, and Appendix B (p. 23) shows loss curves only. A speed claim with no training or data cost is the §5.3 R3 fail case.
P00550	R4	None	partial	partial	Designs are compared with standard-TO compliance (Fig. 13, p. 18) [inferred: fine-mesh FE evaluation], but volume-constraint satisfaction and failure counts are not reported (only a qualitative no-disconnection remark, Sec. 6.2, p. 16), and the bridge uses a coarse-mesh compliance estimate (Sec. 6.3, p. 17): the §5.3 R4 partial case.
P00550	claimed speed-up	None	26.5	30	The primary outcome is the abstract's claim (p. 1), up to 30 times faster, so its stated factor 30 is entered. The largest measured value is 26.5 (Fig. 16, p. 20: 39 h vs 1.5 h at 10.6 million elements, NB = 3; 26.0 from the rounded times [derived]), which the basis records.
P00610	access	no full text	full text	full text	The second coder's legal route worked for the adjudicator: NSF PAR, the funder repository named by the acknowledgments' NSF award (par.nsf.gov/servlets/purl/10521324), serves the CC BY-NC-ND version of record (12 pp.) with no bot check. The first coder had not tried this route.
P00610	stratum	None	speed	speed	The abstract (p. 1) compares the CNN's 15 ms prediction with the finite-volume simulation that each optimization iteration otherwise needs (3 to 5 orders of magnitude faster), a quantitative cost-proxy comparison (§5.1 rule 1), so the stratum is speed, as for P00149.
P00610	R1	None	pass	pass	At the per-evaluation level of the primary outcome the CNN is validated against the same NHT solver it replaces (test MAPE 0.35-2.35% on 26,000 images per ratio, Table 5, p. 7), so the comparison sits within its reported error (§5.3 R1 pass, as in P00149). The optimization runs share optimizer and budget and are re-checked by NHT at every accepted swap (Sec. 5, Figs. 7-10).
P00610	R2	None	partial	partial	The NHT baseline solves a 16,384-cell elliptic problem by iterated TDMA line sweeps, needing 32,903-154,307 iterations and 34.2-371.2 s per image (Sec. 2.3, Table 1, p. 4): neither a sparse direct nor a multigrid-Krylov solver, so the implementation looks inefficient without a direct comparison (§5.3 R2 partial). (c) does not apply below 10^5 DOFs, and (a)/(e) need a re-test.
P00610	R3	None	pass	pass	Data generation (520,000 NHT runs with per-image times, Sec. 2.3 and Table 1, p. 4) and training (about 1 h on 4 V100 GPUs, Sec. 4.2, p. 7) are both reported, and the speed claims are explicitly per image or per iteration (abstract; Sec. 5.1-5.4), which is the §5.3 R3 pass case.
P00610	R4	None	partial	pass	All four returned designs (and every accepted swap) are re-evaluated by the NHT finite-volume model (Figs. 7-10, pp. 8-9), porosity is held exactly by the one-for-one swap rule (Sec. 5, p. 7), and all runs are reported. The disconnected-material check the second coder asks for is not a constraint of this two-phase conduction problem (Sec. 5.4 treats a checkerboard as admissible) [judgment], so unsure means pass.
P00610	claimed speed-up	None	5473	2280	The abstract's comparison (15 ms against the finite-volume simulation, p. 1) spans 2,280-24,747 with the Table 1 times (p. 4) [derived], and the project enters the lower bound of a stated range (as in P00140 and P00549); the article itself states 2,280x for k_s = 10 (Sec. 5.1, p. 8). The second coder's 5,473 is a median of CPU-based ratios (20 ms, Sec. 4.2), a statistic the article does not use.
P00698	access	full text	no full text	full text	The first coder's route worked for the adjudicator: MDPI's own file host (mdpi-res.com) serves the CC BY version-of-record PDF (30 pp.) with no bot check, a route PROTOCOL §11 accepts. The second coder tried www.mdpi.com (article page and PDF path, both an Akamai block) and a Wayback snapshot of the article page (no article body), not the file host.

Record	Item	First	Second	Decision	Reason
P00698	stratum	speed	None	speed	The abstract (p. 1) compares the FE share of run time, over 90% for SIMP against 40% for the network method, and presents it as lower computational cost; under §5.1 rule 1 this cost-proxy comparison is the primary outcome, so the stratum is speed. The 40% matches no FE share in Fig. 14 (17-65%, p. 19), which is recorded but does not change the stratum.
P00698	R1	partial	None	partial	Only run-time shares are reported (Figs. 13-14, pp. 18-19), not tied to a design case, while quality is reported separately as unlabelled objective values within about ±9% of SIMP (Tables 1-3, pp. 14-15) with no volume fractions, and SIMP stops at a fixed 20 iterations (Sec. 4, p. 10): quality reported, matching loose (§5.3 R1 partial).
P00698	R2	pass	None	pass	The SIMP implementation, solver type and hardware are not stated and no absolute times exist (Sec. 2.1, 5.4), so none of (a)-(e) can be established; the fixed 20 iterations are not twice the 10-20 steps that Sec. 5.1 (p. 12) reports for convergence. Unsure means pass.
P00698	R3	partial	None	partial	Training time, dataset size and break-even are missing; the only cost figure is about 30 min for generating randomly partitioned TO iterations (Sec. 3, p. 7), read as a data cost, and the efficiency claim covers the whole design process (abstract), not per-query inference. A cost reported with no amortization is the §5.3 R3 partial case.
P00698	R4	partial	None	partial	Tables 1-6 give an objective value for every network and SIMP design [inferred: FE compliance; the quantity is unlabelled], but the volume fractions of the network designs are not reported and no failures are counted (Appendix A shows images only): the §5.3 R4 partial case.
P00740	access	full text	no full text	full text	The first coder's route works: the University of Utah SCI Institute repository (publications.sci.utah.edu/publications/Kes2021a/) serves the version-of-record PDF (27 pp., CAD 130 (2021) 102947), an institutional copy. The adjudicator read it in full, so the record is a full-text coding.
P00740	stratum	speed	None	speed	The abstract (p. 1) compares with classical TO only qualitatively ('cost-efficient', 'close to optimal'), so §5.1 rule 2 takes the conclusion's first quantitative comparison: DDNSM 'reduces the computational time to almost half' against standard TO (Sec. 5, p. 24; Table 13, p. 23). That is a time comparison, so the stratum is speed.
P00740	claimed speed-up	1.86	None	1.86	Table 13 (p. 23): 92.00 h for direct SIMP against 49.50 h for the full DDNSM pipeline (dataset 8,836.42 s + training 3,237.20 s + evaluation 19.30 s + warm-started SIMP 166,129.24 s, p. 23), so 1.86 [derived]; the 'without training' row (46.15 h) would give 1.99, within 10 %.
P00740	R1	pass	None	pass	Both arms end with the same SIMP code and stopping rule, the time counts the warm-started refinement (166,129.24 s, p. 23), and the refined designs' compliance errors against the direct designs are reported per sample (Fig. 31, roughly 1e-4 to 5e-2 for Case 6). The unstated stopping rule is shared by both arms, so matching is not loose; unsure means pass.
P00740	R2	pass	None	pass	The comparator is the 88-line and Liu-Tovar MATLAB codes (refs [70], [79]) on a CPU the authors call similar in power to the ML machine, and the warm-started SIMP also runs on CPU (p. 14). At 12,915 DOF (p. 20) (c) cannot apply, (b) and (d) are not observed, and (a)/(e) need the re-test; unsure means pass.
P00740	R3	pass	None	pass	The DDNSM totals include data generation and training (Tables 6, 9, 13, 15; p. 23 lists 8,836.42 s and 3,237.20 s for the 3D case), and break-even counts against direct SIMP are given (Eqs. 19, 28, 30; Table 5). The Eq. 30 arithmetic error (682, not 85 [derived]) is noted but does not remove the accounting.
P00740	R4	pass	None	pass	Every test design is refined by SIMP and its compliance is computed by FE against the direct design (e_Cini, e_Cfin, p. 10; Figs. 10-31); the share of samples where the warm start fails to cut iterations is reported (RoS, Tables 1-4, 8, 10-12, 14). The returned designs are SIMP outputs with the volume constraint enforced [inferred].
P00755	R1	fail	partial	fail	The classical comparator is only an expected time ('will take around 24 h', Sec. 5.2, p. 10); no FE-driven GA run or design is reported, so the ANN designs are never compared with a classical result, only with the IVD targets and with FE re-analysis of themselves (Figs. 9, 12). That is the §5.3 fail marker 'a speed claim is made with no quality comparison'; the partial case presupposes a classical result whose matching is loose.

Record	Item	First	Second	Decision	Reason
P00755	R4	pass	partial	pass	Every returned design is re-analysed by FE ('Optimized true', Sec. 3.3, p. 7; Figs. 9, 12, 14) and all six runs are shown, including the Library A misses (Sec. 5.1, p. 8). The failure modes the rule counts cannot occur here: there is no volume constraint and cubic cells guarantee connectivity (Sec. 3.1.2, p. 4). Only the source of the post-sizing stiffnesses (Fig. 13) is unstated; unsure means pass.
P00786	access	full text	no full text	full text	The open-access article page (academic.oup.com/jcde/article/12/8/127/8203285, CC BY-NC) was served without a challenge, and its PDF link redirects to the publisher's own PDF server (silverchair), which PROTOCOL §11 counts as accessible; the adjudicator read the 27-page PDF in full. The second coder's plain curl request met a Cloudflare challenge and correctly stopped there.
P00786	stratum	speed	None	speed	The abstract (p. 127), introduction (pp. 127-129) and conclusions (p. 150) compare with classical methods only qualitatively, so §5.1 rule 2 takes the first quantitative comparison in the main text: the surrogate is about 500× faster than numerical homogenization (Sec. 2.3, p. 134; Table 2, p. 135). That is a cost comparison, accepted as a proxy as in P00149, so the stratum is speed.
P00786	claimed speed-up	500	None	500	Sec. 2.3 (p. 134) states 'approximately 500 times faster'; the per-example ratios 510-570× match Table 2 (e.g. 0.3190 s / 6.251×10^{-4} s = 510 [derived]), and their mean, 534 [derived], is within 10 % of 500.
P00786	R1	partial	None	pass	Time and accuracy are measured on the same five microstructures, and the surrogate lies within its reported error of the HC result (MAE 0.0011-0.0066, Table 2, p. 135), the §5.3 pass condition at the level of the per-evaluation claim, as decided for P00149. The first coder's reasons (no classical multiscale TO run; illustrative examples, possibly on the database grid) concern the claim's scope and the representativeness of the check, which R4 covers; unsure means pass.
P00786	R2	pass	None	pass	HC implementation, solver and hardware are not stated (Sec. 2.3; Table 2); no GPU is used, the microstructure has about 2×10^4 DOFs [derived], the abstract has no baseline number, and no iteration count applies, so markers (b)-(d) cannot be established and (a)/(e) need a re-test; unsure means pass.
P00786	R3	partial	None	partial	Training time is reported (131 min, Sec. 2.3, p. 134), but the cost of homogenizing the 156,000 database microstructures (Sec. 2.3, p. 133) is not, and no break-even is given: the §5.3 R3 partial case (data-generation cost missing).
P00786	R4	partial	None	partial	The surrogate is checked against FE homogenization only on five illustrative microstructures (Table 2, p. 135); design compliances come from the macroscale FE with surrogate-predicted properties, volume is a soft penalty shown only in plots (Fig. 12A), and the unreasonable design of Fig. 14(6) and the isolated boundary members (Sec. 4.1.4, p. 146) are not counted. FE-based checks without failure or constraint counts are the §5.3 R4 partial case.
P00808	access	no full text	full text	full text	The second coder's route works: mdpi-res.com, MDPI's own file host (allowed by PROTOCOL §11), serves the 26-page CC BY version of record with HTTP 200 and no challenge; the adjudicator read it in full. The first coder tried only www.mdpi.com, which serves an Akamai block.
P00808	stratum	None	speed	speed	The abstract (p. 1) and conclusion (Sec. 7, p. 23) compare with traditional methods only qualitatively, and the 10-minute bridge has no classical time. Among the main-text candidates, the article emphasizes the efficiency comparison with SIMP in its dedicated Sec. 4.3 (Table 1, p. 13: 32.55 s vs 61.36 s), which the abstract and conclusion point 2 echo, so §5.1 rules 2 and 4 give a speed outcome.
P00808	claimed speed-up	None	1.885	1.885	Sec. 4.3 (p. 13) states the time falls 'from 61.36 s to 32.55 s' for the SS beam (Table 1), so 61.36 / 32.55 = 1.885 [derived]; the 3D ratio of Table 2 (7.96 [derived]) is an alternative, not the first comparison.
P00808	R1	None	partial	partial	GeNN is faster and has lower compliance (Table 1, p. 13), so no fail marker holds, but SIMP's 400 iterations come with no stated stopping rule, the §5.3 R1 partial example, and neither method's achieved volume fraction is reported while GeNN enforces volume only by a penalty (Eq. 9, p. 8).
P00808	R2	None	partial	pass	None of (a)-(e) is established: the abstract has no baseline number, the SS beam has about 2.6×10^4 DOFs [derived], and no SIMP convergence criterion or plot exists to test (d) (Sec. 4.3, Table 1, p. 13). The second coder's ground, hardware unstated for both methods, is a reporting gap rather than an established unjustified choice, as decided for P00150 and P00176; unsure means pass.

Record	Item	First	Second	Decision	Reason
P00808	R3	None	na	na	GeNN is optimized from a random initialization for each problem with no dataset or offline training (Sec. 3.1 and 3.3, pp. 6-9), the §5.3 R3 n/a case for per-problem learning.
P00808	R4	None	partial	partial	Compliance of the returned density field is computed by FE for both methods (Sec. 3.2, Eq. 8, p. 8), but Tables 1-2 report no achieved volume fraction and no failure or infeasibility count (only the Sec. 5.2 bridge reports volumes, p. 17): the §5.3 R4 partial case.
P00890	R2	partial	pass	pass	Both arms use the same TO code per run and the claim is a run count (p. 7), so solver and hardware cancel, and the article states none of them; no marker (a)-(e) and no listed partial case (inefficient-looking implementation, unjustified hardware) is established, so unsure means pass. The first coder's ground, that a bracketing search over p could need fewer runs than the uniform 12-point grid (p. 10 expects s to fall with p) [inferred], concerns the search strategy, which §5.3 R2 does not cover; it is kept as a note in the evidence.
P00890	R4	partial	pass	partial	The returned designs come from real TO runs and the predictor is checked against actual ESM runs on 720 held-out samples (R^2 0.96, RMSE 0.027, p. 7), but the crash and hood transfers are graphical only (Figs. 16, 19), and misses of the target similarity, including the non-monotonic predictions near the feature bounds (p. 10), are not counted. FE-based checks without a failure count are the §5.3 R4 partial case.
P00918	access	no full text	full text	full text	The second coder's route works: the UPV institutional repository RiuNet serves the CC BY version of record (bitstream 2f0f92f2-6cb1-4bbd-a15c-168698bb817f, 24 pp.) with HTTP 200 and no challenge; the adjudicator read it in full. The first coder tried only ScienceDirect (captcha) and the HAL notice.
P00918	stratum	None	speed	speed	The abstract (p. 1) and conclusions (Sec. 6, pp. 21-23) claim reduced computation time only qualitatively, so §5.1 rule 2 goes to the main text, whose quantitative comparison with the standard SIMP is the iteration-count acceleration ratio (Eqs. 6-7, p. 13; Table 5, p. 17). Fewer SIMP iterations are a cost proxy, so the stratum is speed.
P00918	claimed speed-up	None	2.67	2.67	Table 5 (p. 17) gives 2.67 for the energy-based metric with Dataset Index 3 and augmentation, the configuration the conclusions recommend (p. 22); the highest cell, 2.86 (L2), is within 10 %. It is an averaged iteration ratio, not a time.
P00918	R1	None	partial	partial	Both arms converge with the same SIMP code, so no fail marker holds, but the convergence criterion is not stated numerically (Sec. 2.2, p. 4) and final compliance is compared only for two example cells and qualitatively for the assembled structure (Figs. 29-31, pp. 20-23): the §5.3 R1 partial case.
P00918	R2	None	pass	pass	Both arms use the same FE/SIMP code and differ only in the initial guess (Sec. 3.3, Fig. 10, pp. 9-11), and the claim is an iteration ratio, so solver and hardware cancel; the abstract has no baseline number and no marker (a)-(e) is established; unsure means pass.
P00918	R3	None	fail	partial	There is no training phase (Sec. 3.3, p. 10), and the data cost is reported as sample counts, with each of the 984 context-based samples needing its own cell TO (Tables 1-2, pp. 7-9; p. 22), but never in time or iterations and without break-even. That is 'costs reported with no amortization' (§5.3 R3 partial), not a speed claim with no data cost at all; the second coder's supporting quotation does not occur in the text.
P00918	R4	None	partial	partial	Returned cells are real SIMP outputs, but their compliance is shown only for two example cells (Figs. 29-30) and qualitatively for the assembly (Fig. 31), and no infeasibility or failure count is given (Sec. 5.2-5.3, p. 20): the §5.3 R4 partial case.
P01217	R1	fail	partial	fail	DL inference (3.004 s) is timed against complete SIMP and Abaqus runs (Table 4, p. 35), while every DL design uses more material (VF 0.53-0.54 vs SIMP 0.49-0.51, Table 3, p. 34) and has higher strain energy than SIMP and Abaqus (Fig. 10, p. 35). The 8% VF deviation (p. 34) exceeds the 5% error the abstract reports, and no matching is attempted: the §5.3 fail marker, not loose matching.
P01217	R4	fail	partial	partial	The primary comparison's five DL designs get a measured VF (Table 3, p. 34) and a strain energy that needs an analysis of each returned design (Fig. 10, p. 35; FE inferred). The five cases are the designated comparison set, not outputs picked from a larger pool, so the selected-examples fail marker does not clearly apply. Failures are not counted: the §5.3 R4 partial case.

Record	Item	First	Second	Decision	Reason
P01517	R2	partial	pass	partial	The in-house MATLAB baseline needs about 12.1 s per iteration at 10,712 DOFs (Table 1: 2540.44 s for 210 IGA evaluations, Sec. 5.1) [derived], and Sec. 4 names the sensitivity step, which the FNO replaces, as the bottleneck. The implementation thus looks inefficient while no direct comparison is possible, the §5.3 R2 partial case; shared hardware settles only the hardware criteria.
P01974	stratum	speed	quality	speed	Abstract (p. 1): the quantitative SIMP comparison (87.5 % stiffness, 94.3 % thermal) is followed by the statement that these results need no repeated FE simulations, claimed as a significant cost reduction; §3.2 lists such FE-evaluation savings as a cost proxy. So the §5.1 rule-1 primary outcome is a quality-for-cost comparison, not quality only, as the title and Sec. 1 (p. 6) also frame it.
P01974	R1	fail	partial	fail	The cost claim (abstract, p. 1; Sec. 4.2, p. 22) is made for LLM designs that are measurably worse than the SIMP reference, 12.5 % less stiff (Sec. 4.1 and Fig. 7, pp. 19-20), with no equal-quality or equal-budget matching and no volume fractions. That is the §5.3 R1 fail case, not loose matching of comparable designs.
P01974	R2	partial	na	partial	The stratum is speed, so R2 applies. SIMP's implementation, solver and hardware are unstated, and the only classical cost, a generic 5 s per iteration (Sec. 4.2, p. 22), is about 1,000× our 4.5 ms PARDISO per-iteration time at 64×64 [inferred]; as it is not a timed run at a stated size and hardware, (e) cannot apply and the inefficient-implementation partial case holds.
P01974	R4	partial	fail	partial	Secs. 4.1-4.2 (pp. 19-22) never say how the 0.875 and 0.9432 ratios were computed, and nothing establishes that they are the heuristic Q(X) scores (Fig. 6's objective plateau, about 0.91, differs from 0.875), so the surrogate fail case is not established. Volume fractions and failure counts are unreported, the §5.3 R4 partial case.
P01974	claimed speed-up	3.33	None	3.33	No factor is stated. The only printed classical and ML times are about 1.5 s per LLM-DSTO iteration against the author's 5 s per iteration for traditional TO (Sec. 4.2, p. 22), so 5/1.5 = 3.33 per iteration [derived]; it quantifies the speed-stratum primary outcome, and the basis notes that the 5 s is a generic figure, not a timed SIMP run.
P02187	R2	partial	pass	partial	Same-CPU timing settles only the hardware criteria. Table 2 (p. 97) implies about 1.25 s per SIMP iteration at about 2.4×10^4 DOFs [derived], and one FDM solve at 160,000 voxels takes almost 5 min (p. 96), with the solver unnamed; DL4TO's source (outside the article) uses SciPy's general sparse LU. The implementation thus looks inefficient while no 3D re-test allows a direct comparison, the §5.3 R2 partial case (as in P01517).

7. Access judgement calls

Of the 200 sampled records, 90 were listed as paywalled and skipped. Of the other 110, 83 had a readable legal full text, 23 offered only an abstract and 4 were blocked by a publisher bot check. Rules applied (protocol §11): an open-access copy on the publisher's own file host counts as accessible; no CAPTCHA or bot check was bypassed; an article without a readable full text keeps its screening decision (include, unclear) and enters no rate. The records below are the judgement calls: each shows what could be read.

Record	Article	Access	What was read
P00533	Jeong et al. (2023)	abstract only	none: abstract only (OpenAlex abstract as stored in the pool); full text not reachable
P00537	Li et al. (2019)	abstract only	none: abstract only, read on the LSU institutional repository record page (repository.lsu.edu/eecs_pubs/698), which states the document itself is not available there; the pool record has no abstract
P00597	Sun et al. (2022)	abstract only	none; only the abstract was read (pool record, from OpenAlex)
P00601	Gavris and Sun (2024)	no access	none: no accessible full text or abstract
P00605	Liu et al. (2023)	abstract only	none: only the abstract was read (pool record from OpenAlex). The version of record could not be retrieved
P00624	Rade et al. (2023)	no access	none: no full text and no abstract reachable (the pool record from OpenAlex has no abstract; Crossref and Semantic Scholar hold none)
P00648	Brown et al. (2022)	abstract only	none: abstract only (OpenAlex abstract as stored in the pool); the journal full text was not reachable by the first coder or by the adjudicator (2026-09-24) without passing a bot check
P00707	Xing and Tong (2025)	abstract only	none; only the abstract was read (pool record, from OpenAlex)
P00709	Gao et al. (2025)	abstract only	none: only the abstract was read (pool record from OpenAlex). The version of record (ScienceDirect PII S0888327025013718) could not be retrieved by script; the adjudicator's own attempts on 2026-09-24 (article page and its pdfft PDF link) also met a captcha page, which was not bypassed

Record	Article	Access	What was read
P00712	Kallioras and Lagaros (2020)	abstract only	none: abstract only (OpenAlex abstract as stored in the pool); full text not reachable
P00822	Pereira and Driemeier (2024)	abstract only	none: abstract only (OpenAlex abstract as stored in the pool); full text not reachable by this agent
P00860	Garrelts et al. (2021)	abstract only	none: only the abstract was read (pool record from OpenAlex). Neither the version of record nor the Fraunhofer repository copy could be retrieved
P00875	Ma et al. (2023)	abstract only	none: only the abstract was read (pool record from OpenAlex). The version of record could not be retrieved
P00983	Xing et al. (2025)	abstract only	none; only the abstract and highlights were read (pool record and OpenAlex metadata)
P01016	Ghansiyal et al. (2024)	abstract only	none; only the abstract was read (pool record from OpenAlex). The version of record could not be retrieved
P01151	Honarvar et al. (2026)	no access	none; neither the full text nor an abstract could be read (OpenAlex and Crossref hold no abstract)
P01268	Aroyan et al. (2026)	abstract only	none; only the abstract was read (pool record, OpenAlex and DOAJ metadata)
P01289	Vulimiri et al. (2026)	abstract only	none: only the abstract and highlights were read (pool record from OpenAlex). The version of record (ScienceDirect PII S2214860426000205) could not be retrieved by script
P01390	Venugopal and Anand (2023)	abstract only	none: abstract only (OpenAlex abstract as stored in the pool); full text not reachable by this agent
P01827	Carson and Hirschfeld (2026)	abstract only	none: only the abstract was read, on the publisher's article page (jhsr.net/index.php/jhsr/article/view/72); the page offers no full-text galley
P01870	Chu et al. (2024)	abstract only	none; only the abstract was read (pool record from OpenAlex; the DOAJ API record repeats it). The version of record could not be retrieved
P01907	Ignuta-Ciuncanu et al. (2025)	abstract only	none: abstract only (OpenAlex abstract as stored in the pool); full text not reachable
P01994	Tskhondia (2025)	no access	none; neither the full text nor an abstract could be read (DataCite and OpenAlex hold no abstract)
P02084	Ignuta-Ciuncanu et al. (2026)	abstract only	none; only the abstract was read (pool record and OpenAlex metadata)
P02128	Jadhav et al. (2024)	abstract only	none: only the abstract was read (pool record from OpenAlex). The version of record could not be retrieved by script
P02296	Liu et al. (2026)	abstract only	none: abstract only (OpenAlex abstract as stored in the pool); full text not reachable
P02479	Garland et al. (2021)	abstract only	none: abstract only (OpenAlex abstract as stored in the pool); full text not reachable

8. Rule markers behind Sect. 4.7

Markers recorded with the speed-stratum verdicts in the coding records; a verdict may carry more than one marker, and markers flagged [inferred] are counted with their base marker.

Rule	Verdict	Marker	Count
R1	fail	R1-unmatched	7
R1	fail	R1-no-quality (image-only)	1
R1	fail	R1-no-quality	1
R1	partial	R1-loose	14
R1	pass	R1-matched-per-evaluation	3
R1	pass	R1-matched	3
R2	partial	R2-inefficient-impl	9
R2	partial	R2-hardware-unjustified	2
R2	partial	R2-inefficient-implementation	1
R2	partial	R2(e) re-test pointer	1
R2	partial	R2(e) re-test candidate	1
R2	pass	R2-none-established	15
R2	pass	R2-gpu-baseline	1
R2	pass	R2-pass-shared-solver	1
R3	fail	R3-no-cost	6
R3	fail	R3-no-cost-reported	1
R3	partial	R3-no-data-cost	7
R3	partial	R3-no-amortization	5
R3	partial	R3-no-training-cost	4
R3	pass	R3-full-accounting	3
R3	pass	R3-full-cost	1
R3	pass	R3-costs-reported	1
R3	pass	per-query framing	1
R3	na	R3-na-per-problem	2
R3	na	R3-na-per-problem-learning	1
R4	fail	R4-image-metric	3
R4	fail	R4-selected-examples	1
R4	partial	R4-no-failure-rate	13
R4	partial	R4-surrogate-predicted	1
R4	partial	R4-partial-verification	1
R4	partial	R4-no-constraint-report	1

Rule	Verdict	Marker	Count
R4	pass	R4-FE-verified	8

9. Post hoc cross-provider coding check

This check was added after the main analysis, because no independent human coder was available. Two OpenAI models, GPT-6-Sol and GPT-6-Luna, coded the same 110 records from the inputs of the blind second pass: the protocol with masked record identifiers, the coding schema, the code lists and the bibliographic queue lines. They ran in the Codex command-line tool (0.158.0) with reasoning effort high, live web search, a workspace-write sandbox with network access, and memories, plugins, browser use and subagents switched off. Each run held five or six records in a folder outside the project; the models saw no earlier coding and not each other's records. A scan of every run's event log found no access to project, adjudication or earlier-coding files, and every run log reported no contact with study material. Full texts came from legitimate open sources (the download helper refused shadow libraries), and the run logs report no bypassed bot check. Thirty-nine GPT-6-Luna records failed schema validation (values outside the allowed reporting codes, a missing stage field, two missing objects on excluded records); new runs of the same model repaired the format only, and a check confirmed that access, inclusion, stratum and all rule verdicts were unchanged. Both models also judged the 17 re-screen referrals of section 5 at full text.

Reproduction identifiers for the two model APIs:

Model	API identifier
GPT-6-Sol	`gpt-6-sol`
GPT-6-Luna	`gpt-6-luna`

Agreement uses the same Cohen's κ and bootstrap as section 3 (2,000 resamples, seed 20260924); rule items count records that both coders included.

Pair	Item	Pairs	Raw agreement	κ	95 % CI
Claude first vs second pass	Inclusion	110	88 %	0.62	0.42 to 0.80
Claude first vs second pass	Stratum	110	70 %	0.53	0.40 to 0.65
Claude first vs second pass	R1	82	55 %	0.35	0.21 to 0.49
Claude first vs second pass	R2	82	56 %	0.28	0.12 to 0.43
Claude first vs second pass	R3	82	83 %	0.66	0.49 to 0.80
Claude first vs second pass	R4	82	70 %	0.56	0.42 to 0.70
GPT-6-Sol vs Claude first pass	Inclusion	110	85 %	0.56	0.36 to 0.73
GPT-6-Sol vs Claude first pass	Stratum	110	75 %	0.60	0.48 to 0.72
GPT-6-Sol vs Claude first pass	R1	79	43 %	0.16	0.03 to 0.30
GPT-6-Sol vs Claude first pass	R2	79	58 %	0.30	0.13 to 0.47
GPT-6-Sol vs Claude first pass	R3	79	80 %	0.56	0.36 to 0.73
GPT-6-Sol vs Claude first pass	R4	79	70 %	0.56	0.41 to 0.69
GPT-6-Sol vs Claude second pass	Inclusion	110	85 %	0.51	0.29 to 0.70
GPT-6-Sol vs Claude second pass	Stratum	110	72 %	0.57	0.42 to 0.69
GPT-6-Sol vs Claude second pass	R1	80	44 %	0.20	0.07 to 0.32
GPT-6-Sol vs Claude second pass	R2	80	64 %	0.39	0.21 to 0.55
GPT-6-Sol vs Claude second pass	R3	80	74 %	0.50	0.32 to 0.66
GPT-6-Sol vs Claude second pass	R4	80	69 %	0.55	0.41 to 0.69
GPT-6-Luna vs Claude first pass	Inclusion	110	85 %	0.48	0.26 to 0.66
GPT-6-Luna vs Claude first pass	Stratum	110	65 %	0.46	0.33 to 0.59
GPT-6-Luna vs Claude first pass	R1	84	49 %	0.25	0.12 to 0.39
GPT-6-Luna vs Claude first pass	R2	84	58 %	0.31	0.16 to 0.47
GPT-6-Luna vs Claude first pass	R3	84	73 %	0.46	0.29 to 0.63
GPT-6-Luna vs Claude first pass	R4	84	62 %	0.44	0.29 to 0.58
GPT-6-Luna vs Claude second pass	Inclusion	110	88 %	0.54	0.30 to 0.75
GPT-6-Luna vs Claude second pass	Stratum	110	68 %	0.50	0.36 to 0.64
GPT-6-Luna vs Claude second pass	R1	87	44 %	0.20	0.07 to 0.33
GPT-6-Luna vs Claude second pass	R2	87	53 %	0.23	0.07 to 0.38
GPT-6-Luna vs Claude second pass	R3	87	67 %	0.42	0.26 to 0.57
GPT-6-Luna vs Claude second pass	R4	87	62 %	0.45	0.30 to 0.59
GPT-6-Sol vs GPT-6-Luna	Inclusion	110	89 %	0.61	0.39 to 0.79
GPT-6-Sol vs GPT-6-Luna	Stratum	110	74 %	0.59	0.47 to 0.71
GPT-6-Sol vs GPT-6-Luna	R1	86	56 %	0.23	0.11 to 0.36
GPT-6-Sol vs GPT-6-Luna	R2	86	55 %	0.26	0.11 to 0.41
GPT-6-Sol vs GPT-6-Luna	R3	86	72 %	0.47	0.30 to 0.63
GPT-6-Sol vs GPT-6-Luna	R4	86	81 %	0.71	0.58 to 0.83

Headline counts with each model's own inclusion, stratum and verdicts (adjudicated: 8/28 (29 %), 8/28 (29 %), 24/28 (86 %) and 15/28 (54 %)).

Coder	R1 fail	R1-or-R2 fail	R1-or-R2 fail or partial	R2 pass by default
GPT-6-Sol (own speed claims)	21/32 (66 %)	22/32 (69 %)	32/32 (100 %)	16/32 (50 %)
GPT-6-Luna (own speed claims)	14/38 (37 %)	14/38 (37 %)	32/38 (84 %)	25/38 (66 %)

Re-screen referrals at full text (17 records). Eligible records are those included at full text that also pass the draw rules; the false-exclusion rate divides them by the 188 re-screened exclusions.

Coder	Included	At full text	Eligible	False-exclusion rate (95 % CI)
GPT-6-Sol	6	4	4	2.1 % (0.6 to 5.4 %)
GPT-6-Luna	5	3	3	1.6 % (0.3 to 4.6 %)